

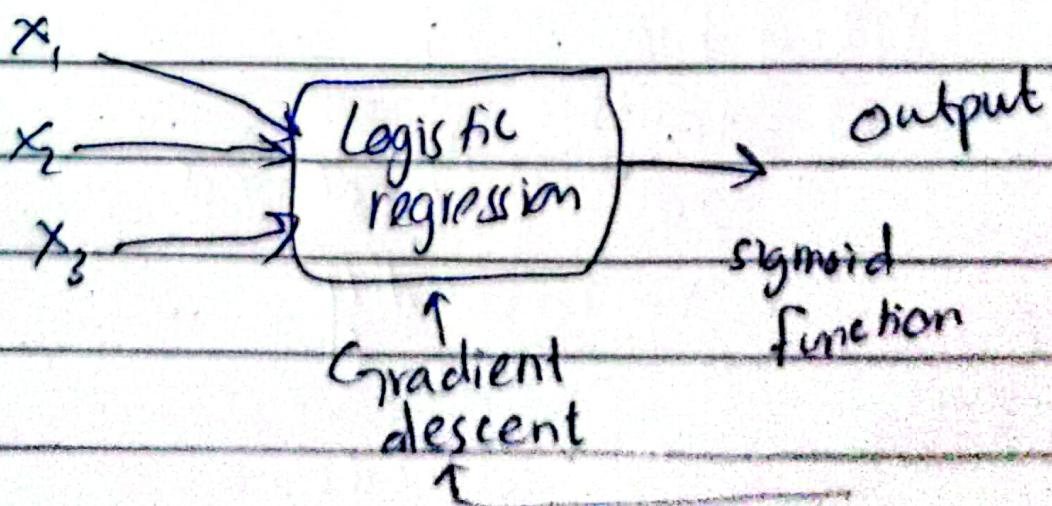
manual way

Logistic Regression: (scratch)

Standard scalar: sab features ko aik scale pr ly aana taaky biasness na aye. saary data ka mean aur standard deviation same rakhy ga by that, saary data ko aik hi scale mein rakhy ga

→ split data into training and testing.

Algorithm → weights
biased



loops vs epoch

epoch $\uparrow$ loss $\downarrow$ $\rightarrow$ before learning.

Hyper parameters $\rightarrow$ jo algorithm ko ^{learning} ~~change~~ design kony sy pehly change krlo.

parameter flow $\rightarrow$ parameters changes during learning

sigmoid function: converts ^{linear equation} learning into values

0 to 1 format.

fit function: use for training

Polynomial regression by sklearn.

Logistic regression

Step 1: weight

har feature ka aik weight

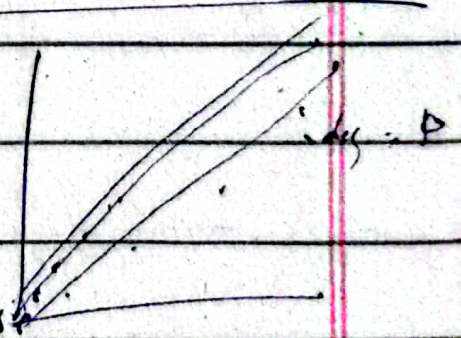
hoga aur usy zero sy initialize krein gy.

Step 4: threshold probability ≥ 0.5

Step 2: calculate

$Z = \text{features} \times \text{weight} \rightarrow$ then add

Step 3: sigmoid $\rightarrow$ new probability



Gradient descent:

model ko kesi data chaly ge
weights correct hain:

$$\text{error} = \text{prediction} - \text{Actual}$$

weights ko kis direction mein
aur kitna change karna hy.